MIL-P-50088A
9 SEPTEMBER 1960

SUPERSEDING
MIL-P-50088 (CmlC)
10 SEPTEMBER 1959

MILITARY SPECIFICATION

PURIFIER, AIR, M2A1

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers a filter unit designed to remove toxic aerosols and gases from atmospheric air at the rate of 12 cubic feet per minute.

2. APPLICABLE DOCUMENTS

2.1 **Government documents.** The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

UU-T-111 — Tape; Paper, Gummed (Sealing and Securing).

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Boxes, Fiber.

PPP-T-60 — Tape, Pressure-Sensitive Adhesive, Waterproof for Packaging and Sealing.

MILITARY

MIL-B-131 — Barrier Material; Water Vaporproof, Flexible.

MIL-P-116 — Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-282 — Filter Units, Protective Clothing, Gas Mask Components and Related Products; Performance Test Methods.

DRAWINGS

CHEMICAL CORPS

D5-19-2263 — Purifier, Air, M2A1; Bill of Material.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

FSC 4240

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risks of reasons of security or prisoner safety (e.g., sulfide

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies of these freight classification rules should be addressed to Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 *Materials.* All materials cited on Drawing D5–19–2263 or on subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 *Components.* All components of the item shall conform to the specifications and drawings listed on Drawing D5–19–2263.

3.2 Manufacture and assembly. The purifier shall be manufactured and assembled as shown on Drawing D5–19–2262, which is listed on Drawing D5–19–2263.

3.3 Performance. When tested in accordance with 4.5.1, the air purifier shall deliver a total air output of 12.0 + 1.5 cubic feet per minute (cfm) against a static pressure of 5.9 inches of water.

3.4 Power consumption. When tested in accordance with 4.5.1 on direct current, the air purifier shall not draw in excess of 5 amperes.

3.5 Resistance to rough handling. The air purifier shall show no indication of damage and shall meet the requirements of 3.3 after treatment in accordance with 4.5.2.

3.6 Workmanship. Poor workmanship shall not be allowed. Damage, such as bent, chipped, or burred metal sections, and contamination by foreign, oily, or viscous material shall be indicative of poor workmanship.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Objective evidence. The supplier shall present objective evidence as required by the Government representative that all requirements of section 3 and of section 5 have been met (see 6.2).

4.3 Alternative inspection (including testing) procedures. The supplier may utilize any alternative inspection procedure which will assure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to its institution. In case of dispute, the procedures of this specification will govern.

4.4 Inspection provisions.

4.4.1 *Lotting.* A lot shall consist of the purifiers manufactured from the same materials by one manufacturer, in one plant, and offered for acceptance at one time. No lot of purifiers shall contain more than one lot of gas filters or one lot of particulate filters.

4.4.2 *Sampling.*

4.4.2.1 *Nondestructive examination.* Sampling shall be conducted in accordance with Standard MIL–STD–105.

4.4.2.2 *Resistance to rough handling.* Eighteen purifiers shall be taken from a lot. Six of the purifiers shall be used as a first

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sample and 12 shall be used as a second sample for testing in accordance with 4.5.2.

4.4.3 *Examination.* Sample purifiers and the levels A and B preparation for delivery thereof shall be examined in accordance with the classification of defects and with Standard MIL–STD–105.

4.4.4 *Classification of defects.*

4.4.4.1 *Purifier, air, M2A1* (Drawing D5–19–2262 and subsidiary drawings, which are listed on Drawing D5–19–2263).

Categories	*Defects*	*Acceptance standards*
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Component missing, incorrect, or incorrectly assembled.	
102	Location for nuts (2) incorrect (Drawing D5–19–881).	
103	Location for floating anchor nut, right angle in upper and lower half of body incorrect	
104	Basic dimensions incorrect (Drawing C5–19–2265).	
105	Body dimensions incorrect.	
106	Location of holes in upper and lower half of body incorrect.	
107	Location of bulk head incorrect (Drawing D5–19–881).	
Minor:	AQL 2.5 percent defective	
201	Protective coating missing.	
202	Workmanship unsatisfactory. (see 3.6).	
Major A:	AQL 0.1 percent defective	
301	Performance unsatisfactory	4.5.1

4.4.4.2 *Preparation for delivery* (section 5).

Categories	*Defects*
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Containers incorrect.
102	Packaging method incorrect.
103	Container closure incorrect.
104	Container damaged (broken, torn, punctured, or wracked).
105	Strapping loose or missing.
106	Marking incorrect, missing, or illegible.

4.5 Tests.

4.5.1 *Performance.* The performance of this item shall be tested with the Chemical Corps Q97 test equipment in accordance with the instructions for using this equipment.

4.5.2 *Resistance to rough handling.* The air purifier shall be rough handled in accordance with method 105.9 of Standard MIL–STD–282 for 15 minutes. The frequency shall be 200 strokes per minute with an amplitude of $\frac{3}{4} \pm \frac{1}{16}$ inch.

4.6 Acceptance/rejection criteria. If one air purifier fails in the first sample when tested according to 4.5.2, the second sample shall be tested. Failure of a total of two air purifiers from the combined first and second samples shall reject the lot.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Level A.* The air purifiers shall be packaged in accordance with method 1A–14 of Specification MIL–P–116, using class 1 water vaporproof barrier material conforming to Specification MIL–B–131 as the heat-sealable bag material. The inner container shall conform to type I, class 1 of Specification PPP–B–636, and the outer container shall conform to V3c, style RSC of the same specification. The inner box shall be closed and the flaps secured with gummed tape conforming to UU–T–111. The outer container shall be closed by taping all seams, including the manufacturer's joint, with type III, class 1, pressure-sensitive tape, conforming to Specification PPP–T–60, extending the tape over all corners and edges of the box at least three inches.

5.2 Packing. Levels of packing shall be as specified (see 6.1).

5.2.1 *Level A.* Air purifier packaged as specified in 5.1 shall be packed not to exceed 200 pounds gross weight in a class 2, style 2, or 2½, nailed wood box conforming to Specification PPP–B–621. The boxes shall be closed as specified in the applicable requirements of the wood box specification and banded with

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galvanized strapping in accordance with the appendix thereto.

5.2.2 *Level B.* Air purifiers packaged as specified in 5.1 shall be packed as specified in 5.2.1, except that class 1, style 2, or 2½, nailed wood boxes and coated-finish strapping shall be used.

5.2.3 *Level C.* Air purifiers packaged as specified in 5.1 shall be packed in a manner to insure carrier acceptance and safe delivery at first destination for immediate use. Containers shall be in accordance with the Uniform Freight Classification rules or regulations of other carriers applicable to the mode of transportation. Containers shall be of uniform size, shape, and material.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Level of packaging and packing required.

6.2 Objective evidence. Provisions for objective evidence and inspection records, and for maintenance of inspection records will be specified by the procuring activity.

6.3 Federal stock number. The Federal stock number for this item is 4240–307–7805.

6.4 International interest. Certain provisions of this specification are the subject of international standardization agreement. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Chemical Corps
Navy—Marine Corps
International interest (see 6.4)

Preparing activity:
Army—Chemical Corps

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